

Member 2 Report

Data Understanding, EDA & Data Quality

Group Assignment – 2026-AI-08K

House Price Prediction and Valuation Feature Analysis

Student Name	Student ID	Core Responsibility
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1. Objective of Member 2 Work

The purpose of this section is to understand the Ames Housing training dataset, identify important patterns related to the target variable SalePrice, and assess data-quality issues that may affect later preprocessing and model development. The analysis focuses on dataset structure, variable types, missing values, duplicates, target distribution, numerical and categorical relationships, potential outliers, low-variation features, and internal consistency.

2. Dataset Understanding

The training dataset contains 1,460 property records and 81 columns. Each row represents one residential property observation. SalePrice is the target variable to be predicted, while the remaining variables describe property characteristics such as size, quality, location, age, basement, garage, and sale conditions.

Item	Result
Number of rows	1,460
Number of columns	81
Numerical-type columns	38
Categorical/object columns	43
Target variable	SalePrice
Duplicate rows	0

Pandas data types alone do not fully describe the meaning of each feature. Some variables stored as numbers represent categories or ordered ratings; for example, MSSubClass is a building-class code and OverallQual is an ordinal quality rating.

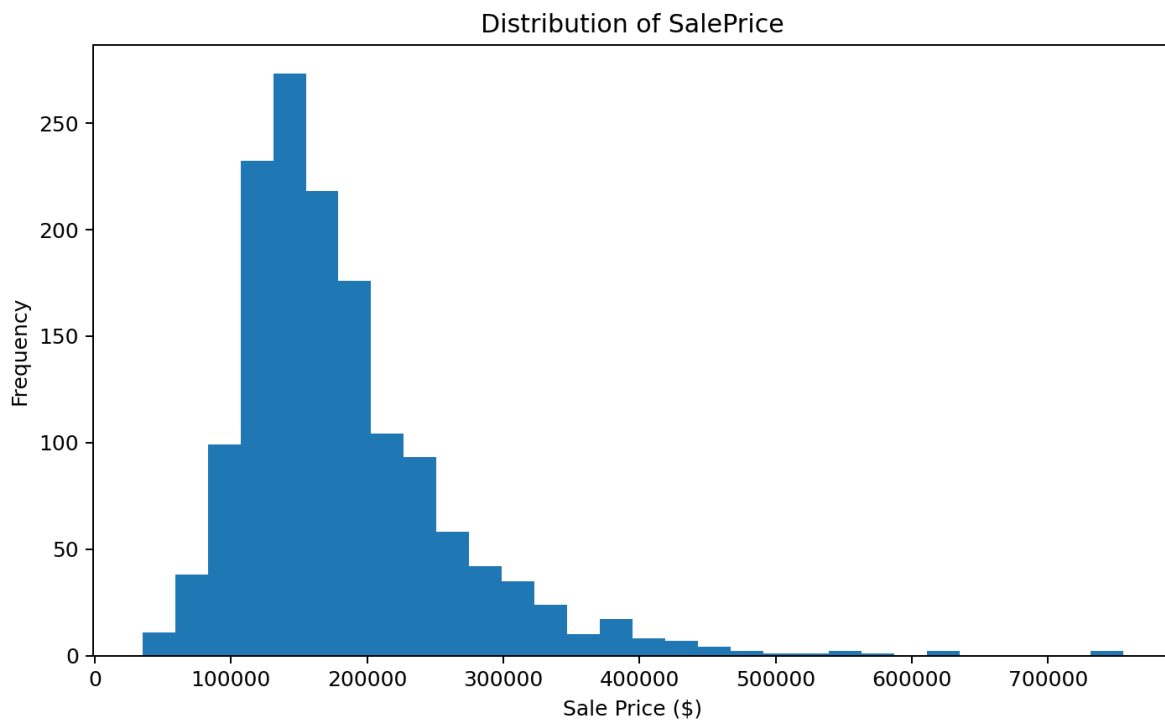
3. Missing-Value Analysis

Several features contain missing values. The highest missingness occurs in features representing optional property characteristics. These missing values should be interpreted carefully because some may represent the absence of a feature rather than unknown or incorrectly recorded data.

Variable	Missing Count	Missing %
PoolQC	1453	99.5%
MiscFeature	1406	96.3%
Alley	1369	93.8%
Fence	1179	80.8%
MasVnrType	872	59.7%
FireplaceQu	690	47.3%
LotFrontage	259	17.7%
GarageType	81	5.5%
GarageYrBlt	81	5.5%
GarageFinish	81	5.5%
GarageQual	81	5.5%
GarageCond	81	5.5%
BsmtExposure	38	2.6%
BsmtFinType2	38	2.6%
BsmtQual	37	2.5%
BsmtCond	37	2.5%
BsmtFinType1	37	2.5%
MasVnrArea	8	0.5%
Electrical	1	0.1%

4. Target Variable: SalePrice

SalePrice contains no missing values. Prices range from \$34,900 to \$755,000. The mean sale price is approximately \$180,921, while the median is \$163,000. Because the mean is higher than the median and the histogram has a long upper tail, the target distribution is right-skewed.

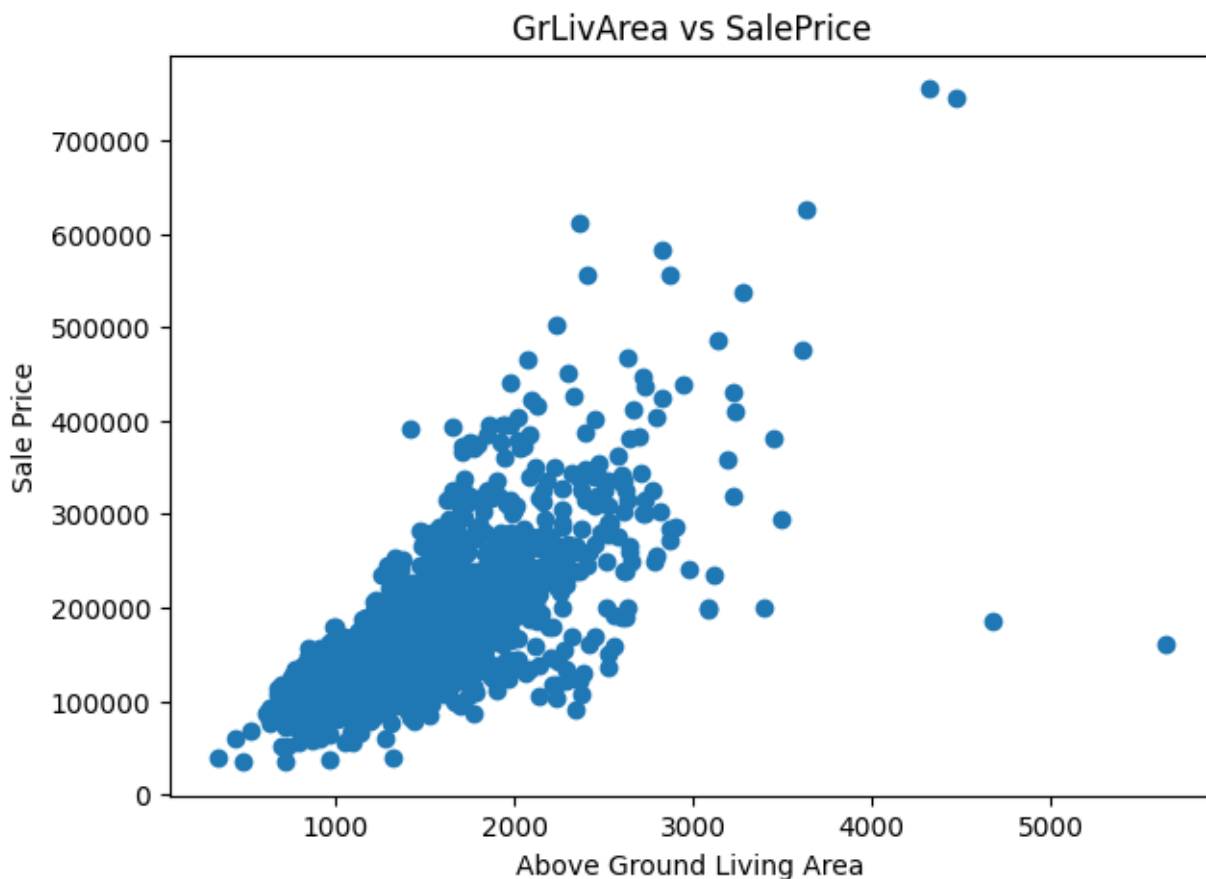


5. Numerical Relationship Analysis

Pearson correlation was used as an exploratory measure for numerical variables. Correlation is interpreted as association rather than causation.

Variable	Correlation with SalePrice
OverallQual	0.791
GrLivArea	0.709
GarageCars	0.640
GarageArea	0.623
TotalBsmntSF	0.614
1stFlrSF	0.606
FullBath	0.561
TotRmsAbvGrd	0.534
YearBuilt	0.523
YearRemodAdd	0.507

OverallQual has the strongest positive numerical relationship with SalePrice ($r = 0.791$), followed by GrLivArea ($r = 0.709$). GarageCars, GarageArea, TotalBsmntSF and 1stFlrSF also show notable positive relationships, suggesting that overall quality, usable living space, garage capacity, basement area and floor area are important variables to investigate during modelling.

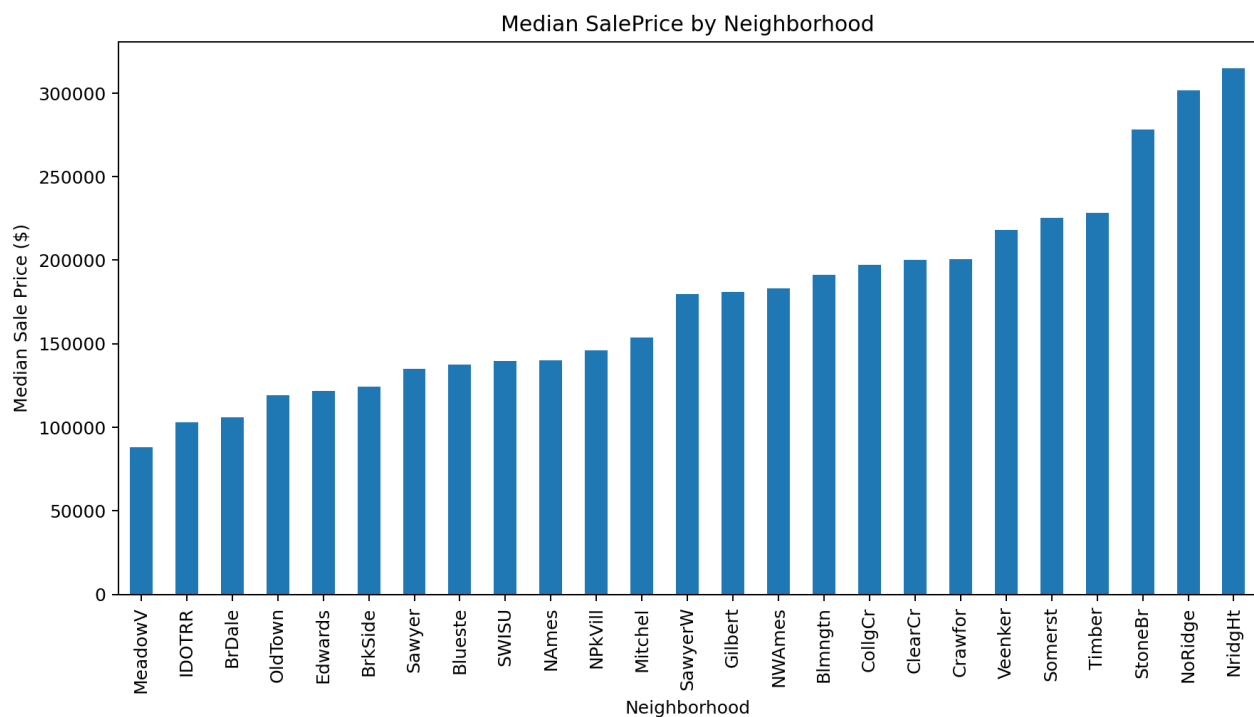


The GrLivArea scatter plot shows a clear positive relationship between above-ground living area and SalePrice. A small number of very large properties do not follow the general pattern and were flagged for outlier investigation.



The OverallQual boxplot shows that median SalePrice generally rises strongly as the quality rating increases. Price variability also becomes larger among higher-quality homes.

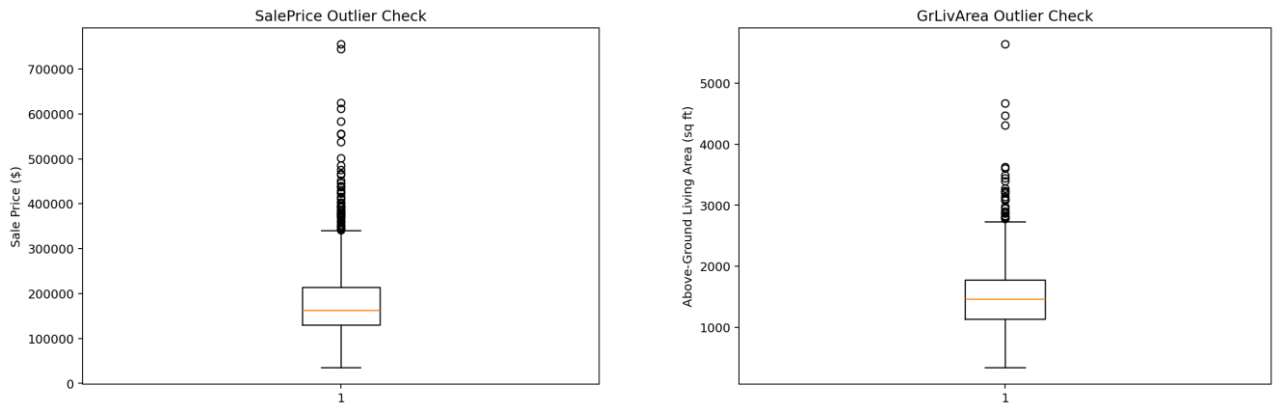
6. Categorical Relationship: Neighborhood



Median sale prices differ substantially across neighborhoods. NridgHt, NoRidge and StoneBr are among the higher-median-price neighborhoods, while MeadowV, IDOTRR and BrDale are among the lower-priced neighborhoods. This indicates that Neighborhood is likely to be an important categorical predictor.

7. Outlier Analysis

Boxplots show high-value SalePrice observations and unusually large GrLivArea values. These observations were flagged rather than removed, because removal is a preprocessing decision that should be justified later.



GrLivArea (sq ft)	SalePrice (\$)
5,642	160,000
4,676	184,750
4,476	745,000
4,316	755,000
3,627	625,000
3,608	475,000
3,493	295,000
3,447	381,000
3,395	200,000
3,279	538,000

Two particularly unusual observations have GrLivArea values of 5,642 and 4,676 square feet but SalePrice values of only \$160,000 and \$184,750. These cases appear inconsistent with the overall positive relationship and should be investigated during preprocessing rather than automatically deleted.

8. Low-Variation / Dominant Features

Some features are dominated by a single value. Such features may contribute limited discriminatory information and should be reviewed during feature selection rather than removed automatically.

Variable	Dominant Value Share
Street	99.59%
Utilities	99.93%
Condition2	98.97%
RoofMatl	98.22%
Heating	97.81%
LowQualFinSF	98.22%
KitchenAbvGr	95.34%

3SsnPorch	98.36%
PoolArea	99.52%
PoolQC	99.52%
MiscFeature	96.30%
MiscVal	96.44%

For example, Street is Pave for approximately 99.59% of properties and Utilities is AllPub for approximately 99.93%. PoolQC and MiscFeature also appear highly dominant partly because missing values reflect the absence of those optional property features.

9. Data Quality and Internal Consistency

The dataset was checked for duplicate rows, structural missingness and consistency among related property features. No duplicate rows were identified.

Garage field	Missing count
GarageType	81
GarageYrBlt	81
GarageFinish	81
GarageCars	0
GarageArea	0
GarageQual	81
GarageCond	81

Basement field	Missing count
BsmtQual	37
BsmtCond	37
BsmtExposure	38
BsmtFinType1	37
BsmtFinType2	38
BsmtFinSF1	0
BsmtFinSF2	0
BsmtUnfSF	0
TotalBsmtSF	0

Where GarageType is missing, the corresponding GarageCars and GarageArea values are zero in the checked records, indicating that the property does not have a garage. Similarly, properties with missing BsmtQual have zero values in the basement square-footage fields, indicating the absence of a basement. Therefore, many missing garage and basement categorical values are structural absence rather than random missing data.

10. EDA Insight Log

Finding	Evidence	Possible Impact
SalePrice is right-skewed	Histogram and summary statistics	A target transformation may be considered later during modelling.
OverallQual strongly relates to SalePrice	$r = 0.791$ and boxplot	Likely an important predictive feature.
GrLivArea strongly relates to SalePrice	$r = 0.709$ and scatter plot	Important size-related feature; also contains unusual observations.
Neighborhood prices differ substantially	Median-price bar chart	Location is likely an important categorical predictor.
Several columns contain missing values	Missing-value table	Missingness requires feature-aware preprocessing.

No duplicate rows found	Duplicate check = 0	No duplicate-row removal is required.
Extreme GrLivArea observations exist	Boxplot and top-area records	Outliers should be investigated before modelling.
Some features are highly dominated by one value	Street, Utilities and other features	Review during feature selection.
Garage/basement missingness is structurally meaningful	Zero area/capacity with missing category fields	Treat feature absence separately from unknown missingness.

11. Data Dictionary Summary

Variable	Meaning	Type	Role
SalePrice	Property sale price in dollars	Numeric target	Target
OverallQual	Overall material and finish quality	Ordinal	Predictor
GrLivArea	Above-ground living area in square feet	Numeric	Predictor
GarageCars	Garage capacity in number of cars	Numeric	Predictor
GarageArea	Garage area in square feet	Numeric	Predictor
TotalBsmntSF	Total basement area in square feet	Numeric	Predictor
1stFlrSF	First-floor area in square feet	Numeric	Predictor
FullBath	Number of full bathrooms above grade	Numeric	Predictor
TotRmsAbvGrd	Total rooms above grade excluding bathrooms	Numeric	Predictor
YearBuilt	Original construction year	Year	Predictor
Neighborhood	Physical location within Ames city limits	Categorical	Predictor
LotArea	Lot size in square feet	Numeric	Predictor
YearRemodAdd	Year of remodel or addition	Year	Predictor
Fireplaces	Number of fireplaces	Numeric	Predictor
BsmntFinSF1	Type 1 finished basement area in square feet	Numeric	Predictor

The full Ames Housing dataset contains 81 variables. The table above summarizes the target and selected important variables highlighted during EDA. The complete official variable descriptions can be retained in the project appendix or supporting documentation.

12. Handover to Member 3

The findings from this analysis should guide the preprocessing and feature-engineering stage. Member 3 should pay particular attention to structural missingness, categorical encoding, possible target transformation, outlier handling, highly dominant variables, and leakage-safe preprocessing. Missing values should not be imputed uniformly without considering the semantic meaning of each feature.